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Multi-domain fake news detection method based on generative adversarial network and graph network

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ABSTRACT

The proliferation of misinformation in today's digital era poses significant challenges, with fake news detection becoming critical to mitigate economic losses and social instability. Despite extensive research efforts, most existing approaches are tailored for single-domain fake news detection, struggling with data distribution discrepancies and domain shifts when applied to multi-domain scenarios. This limitation underscores the urgent need for solutions that address the complexities of cross-domain detection. Here, we propose a novel framework MFGAG that synergistically integrates adversarial networks and graph neural networks with emotional, stylistic, and semantic features to enable precise domain localization. By leveraging these features, the framework effectively models intricate relationships among news articles within the same temporal context, addressing the challenges posed by multi-domain datasets. Experimental evaluations demonstrate that our approach outperforms state-of-the-art methods, achieving an average accuracy improvement of 3.3 percentage points for single-domain news and nearly 1 percentage point for mixed-domain data, culminating in an overall accuracy of 93.1 %. The code involved in this study is publicly available on website https://github.com/SWLee777/MFGAG.

1. Introduction

With the widespread use of social media and online platforms, the speed and reach of information dissemination have increased significantly, and so has the spread of fake news. Fake news not only has a negative impact on social trust and public decision-making but also has the potential to trigger social panic, mislead public opinion, and even influence election results [1]. Consequently, the development of efficient fake news detection methods has become an important topic in the fields of Natural Language Processing (NLP) and Machine Learning. Fake news detection aims to identify misleading reports that deviate from factual information by analyzing news content. This process requires models to have robust text comprehension capabilities, while also considering multi-dimensional information such as the source of the news, its contextual background, and the network features of its dissemination.

Most current fake news detection methods primarily focus on single-domain fake news detection [2]. However, as the scope and formats of news dissemination have diversified, it becomes highly inefficient to first identify the domain and then perform detection. Additionally, these methods struggle to adapt to multi-domain mixed news detection. This limitation arises from significant differences in the characteristics of different news domains. The disparity in the textual meanings across domains hinders the generalization capability of models in cross-domain environments. For instance, the term "virus" in biology and "virus" in computer science represent entirely different concepts. Moreover, single-domain fake news samples are often limited in number, which fails to provide enough training data to support the high-performance models needed for accurate detection, further restricting the performance of these methods.

To address these challenges, researchers have proposed various fake news detection methods, including NLP-based text feature extraction, sentiment analysis, and knowledge graph techniques [3–5]. NLP-based methods evaluate the authenticity of news by analyzing lexical, syntactic, and semantic features within news texts [6]. Although these methods show good performance within a specific domain or dataset, their performance is often unsatisfactory in cross-domain detection tasks due to domain differences.

To improve the performance of cross-domain news detection, some studies have attempted to alleviate the impact of domain differences

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through cross-domain learning models. These models are trained using multi-domain mixed data, enabling them to learn cross-domain features with enhanced generalization capabilities. However, while this approach can somewhat improve the generalization ability of the model, it essentially splits different domains, much like using multiple expert systems to gather domain-specific knowledge and then selecting one expert from among them. This method fails to reveal the interrelationships between features from different domains and can be seen as a parallel version of single-domain detection.

Recently, scholars have proposed true multi-domain fake news detection, which aims to develop detection models that can simultaneously process news content from different domains while capturing information across domains. However, multi-domain detection models still face several challenges in practical applications. Firstly, news from different domains often exhibits significant differences in word frequency distribution, dissemination patterns, and semantic features, making it difficult for models to effectively capture cross-domain feature correlations when handling cross-domain data. This discrepancy exacerbates the complexity of cross-domain detection tasks. Secondly, cross-domain news data frequently suffers from class imbalance, with fake news samples in some domains being scarce. This scarcity hinders the model's ability to accurately learn features from these underrepresented domains. These challenges underscore the need to develop more robust and adaptive multi-domain fake news detection models to comprehensively improve their performance in cross-domain detection tasks.

To achieve true multi-domain information fusion in fake news detection, this paper proposes an innovative framework. First, the BERT model [7] is used for deep information extraction from news texts, combined with an attention mechanism and domain embedding information to perform multi-level feature extraction [8]. Next, adversarial neural networks are introduced to eliminate the impact of domain-specific keywords and emphasize non-keyword information, enabling more precise domain distinction by extracting domain-related information from the text. Subsequently, sentiment analysis is conducted on both the news content and its comments, calculating sentiment differences between them while analyzing the stylistic features of the news to capture variations in news style. Then, a news graph network is constructed based on the textual similarity of news articles, and features are extracted from the structured data using Graph Convolutional Networks(GCN) ([9–11]). This method effectively summarizes both universal and domain-specific information from multi-domain sources. Finally, the textual information, sentiment features, stylistic features, and the features extracted through the GNN are integrated to achieve accurate fake news detection.

The primary contributions of this paper are as follows:

1. **Multi-domain Feature Extraction and Fusion:** This study proposes a multi-level feature extraction method that comprehensively considers sentiment, stylistic, and semantic features. By combining deep learning models, this approach significantly improves the accuracy of cross-domain fake news detection.
2. **Adversarial Training and Attention Mechanism Integration:** By integrating adversarial networks with attention mechanisms, this method effectively addresses the domain shift issue, reduces feature bias caused by certain keywords, and ensures that the model can uncover deeper feature variations across different domains, thereby enhancing fake news detection accuracy.
3. **Modelling Spatial Relationships Using Graph Networks**: This study utilizes Graph Neural Networks (GNN) to model the spatial relationships between multi-domain news articles. By capturing the connections between similar news articles, it achieves true multi-domain fake news detection, rather than a parallel approach to single-domain detection, improving both the accuracy and robustness of fake news identification.

In summary, this paper addresses the challenges of multi-domain fake news detection by proposing a method that integrates BERT, adversarial networks, sentiment analysis, news style analysis, and Graph Neural Networks, providing an effective solution for improving the accuracy and cross-domain generalization capability of fake news detection.

2. Related work

2.1. Content-Based fake news detection

Content feature extraction is a fundamental approach in fake news detection, as it focuses on analyzing the semantic features within news texts to assess their authenticity. Traditional text feature extraction methods, such as the bag-of-words model, TF-IDF, and topic modelling, have been commonly employed for fake news detection [12]. However, these methods often suffer from limitations when applied to cross-domain fake news detection due to the significant differences in text features across various domains, resulting in poor performance. In recent years, deep learning models, particularly those based on BERT [7], have made substantial advancements in text representation learning, significantly improving the ability to capture the underlying semantics of news articles. For example, the BERTGuard model proposed by Alnabhan et al. [4] leverages BERT for feature extraction from multi-domain news texts and incorporates multi-task learning to enhance the detection effectiveness across domains. Additionally, Ding et al. [13] tackled the challenge of cross-event fake news detection by proposing EvolveDetector, a continuous fake news detection method that transfers knowledge between different events at the parameter level without the need to store historical event data. Truică et al. [14] introduced a multi-branch approach, where the content branch focuses on analyzing text context using different word embeddings, while the social branch considers social context information to achieve more robust fake news detection.

2.2. Emotion analysis-based fake news detection

Emotion analysis plays a crucial role in fake news detection, as fake news often contains provocative content designed to elicit strong emotional responses from readers. By analyzing the emotional tone of news texts and associated comments, researchers can gain valuable insights into the authenticity of the news. Existing studies have focused on identifying emotional tendencies within news articles and user comments as a means to assess their veracity [15]. For instance, Liu et al. [16] proposed an emotion-driven detection method that identifies potential misinformation by calculating emotional discrepancies between news texts and user comments. Their approach highlights how emotional analysis can be used to detect inconsistencies or manipulative content. Similarly, this paper incorporates sentiment analysis on both news texts and comments, calculating the emotional differences between the two. By doing so, it enriches the feature set and provides additional insights into the authenticity of the news, thus enhancing the accuracy of fake news detection. This approach not only deepens the understanding of emotional influence on news credibility but also contributes to the development of more robust, emotion-aware models for identifying fake news.

2.3. Knowledge graph-based fake news detection

In recent years, knowledge graphs have become a valuable tool in fake news detection. By establishing relational networks between entities, knowledge graphs provide rich contextual information that helps models assess the authenticity of news content [17]. Xie et al.[18] proposed a joint embedding model that combines structured triples with unstructured entity descriptions for detecting fake news, while Gao et al. [19] introduced a multimodal fake news detection approach that integrates news texts, knowledge concepts, and visual information to generate more comprehensive semantic representations. However, the application of knowledge graphs in multi-domain fake news detection presents several challenges, particularly related to noise introduced during data fusion. Different domains often involve distinct types of entities and relationships, making it difficult to align and integrate knowledge across diverse contexts. This paper addresses these challenges by proposing a method to construct a news network based on similarity thresholds, using Graph Neural Networks (GNN) to improve the detection process. GNN are deep learning models specifically designed to process graph-structured data, which are inherently non-Euclidean in nature. By effectively modeling the complex relationships between nodes, GNNs have demonstrated remarkable analytical capabilities across a wide range of domains. By leveraging GNNs, this approach effectively captures the interconnections between news articles and their related entities across multiple domains, enhancing the model's ability to identify fake news more accurately. This method not only mitigates the issue of data noise but also strengthens the robustness of multi-domain fake news detection. However, there are also some challenges in the integration of graph neural networks, including but not limited to scalability, over-smoothing, adversarial attacks, noisy labels, inductive/conductive learning, etc., which need to be further addressed [20–23].

2.4. Cross-Domain fake news detection

Traditional fake news detection methods have largely focused on single domains, such as politics, health, and entertainment. However, these single-domain models often perform poorly when applied across diverse domains due to the significant variations in textual features and patterns between domains. This limitation has led to growing interest in multi-domain fake news detection as a means to improve detection accuracy and robustness across different news categories. Pennycook et al. (2019) highlighted that many existing models struggle in cross-domain detection, primarily because they fail to adapt to feature variations between domains. In response, several studies have sought to enhance model generalization by incorporating multi-domain learning techniques. For instance, Tong et al. [24] proposed a cross-domain learning model that separately learns features from specific domains and cross-domain embeddings, combining both to better identify fake news across diverse domains.Nan et al. [3] introduced a multi-domain detection model that leverages multiple expert networks, such as TextCNN [25], to capture feature representations from different domains, which improves the accuracy of cross-domain fake news detection. In addition, Wu et al. [26] proposed a clustering framework based on domain and category styles to facilitate style model learning across news domains. They introduced a multi-level contrast loss method to refine style clustering at the domain and category levels, thereby enhancing the generalization and discrimination capabilities of the learned style models. Similarly, Bazmi et al. [27] focused on entities within news articles, using them as key ingredients for learning both domain-invariant and domain-specific representations. This approach addresses the challenges posed by domain shifts and incomplete domain labelling in multi-domain fake news detection.

Generative Adversarial Networks (GANs) constitute a deep learning model composed of two fundamental components: a generator and a discriminator. Through adversarial training, both components are dynamically optimized; the generator endeavors to produce realistic data that can deceive the discriminator, while the discriminator strives to distinguish between genuine and generated data. This iterative process ultimately leads to an equilibrium in which the generated data becomes nearly indistinguishable from the real data. In the context of fake news detection, generative adversarial techniques can be employed to produce increasingly deceptive fake news content, thereby enhancing the model's discriminative capabilities. This approach not only improves the robustness of detection systems but also supports more accurate assessments within the field of journalism.

These efforts represent significant strides towards more effective multi-domain fake news detection. However, challenges remain, such as overcoming the limitations of domain-specific feature learning and improving the integration of cross-domain knowledge. Our research aims to address these issues by proposing a novel framework that integrates multiple domain-specific features and leverages advanced techniques, such as adversarial training, attention mechanisms, and Graph Neural Networks, to enhance the accuracy and robustness of multi-domain fake news detection.

3. Method

3.1. Task definition

The primary objective of this research is to tackle the challenges associated with multi-domain fake news detection, particularly the issues of data distribution discrepancies and domain shifts, which significantly undermine the performance of traditional single-domain models. To address these challenges, this study proposes a novel framework that integrates adversarial networks and graph neural networks with multi-dimensional feature extraction, including emotional, stylistic, and semantic features. The proposed framework aims to achieve precise domain localization, allowing the model to accurately differentiate between domains. Furthermore, it seeks to uncover the complex relationships among news articles within the same temporal context, thereby improving the detection of fake news across diverse domains. The overall goal is to enhance both the accuracy and robustness of fake news detection, ensuring its applicability and effectiveness across a wide range of news domains.

3.2. Problem definition

Let S represent a news article on social media, with the text of the article and its associated comments comprising a different number of tokens (words), denoted as $T = [t_1, t_2, t_3 \ldots\ldots t_l]$, totaling l words. Each article is assigned a true label$\hat{y} \in \{0, 1\}$, where 1 indicates the news is fake and 0 indicates it is real. Additionally, each article is manually categorized by experts into a specific domain, with the domain label denoted as $d = \{d_1, d_2 \ldots\ldots d_N\}$ and N representing the total number of domains. Given a news segment S and its corresponding domain label d, the goal of fake news detection is to ascertain whether the news is fake or real.

Based on the original dataset, we performed further processing to extract preliminary features that facilitate subsequent methods. Furthermore, we assigned a unique ID to each node to prepare for the construction of the graph network.

3.3. Feature processing and extraction

The foundational features are primarily categorized into emotional features, stylistic features, and semantic features. As illustrated in Fig. 1, the basic semantic features utilize RoBERTa [7] for word embeddings. We will provide detailed explanation of the other two types of foundational features.

Emotional features:In our sentiment analysis, we referenced and improved upon the methods from study [28] to process the news and comments in the original texts. First, we calculate the publisher's sentiment, which reflects the subjective emotional stance of the news publisher. For $l$words, we utilize the sentiment lexicon from Dalian University of Technology and a list of synonyms expanded by Harbin Institute of Technology.

Initially, the sentiment categories are denoted as $E = [e_1, e_2, e_3 \ldots\ldots e_c]$, with $c$representing the total number of categories. We compute the sentiment for each word across all categories based on the corresponding entries in our lists, resulting in sentiment values for each word represented as Emo_{t_l}. However, the intensity of sentiment varies among

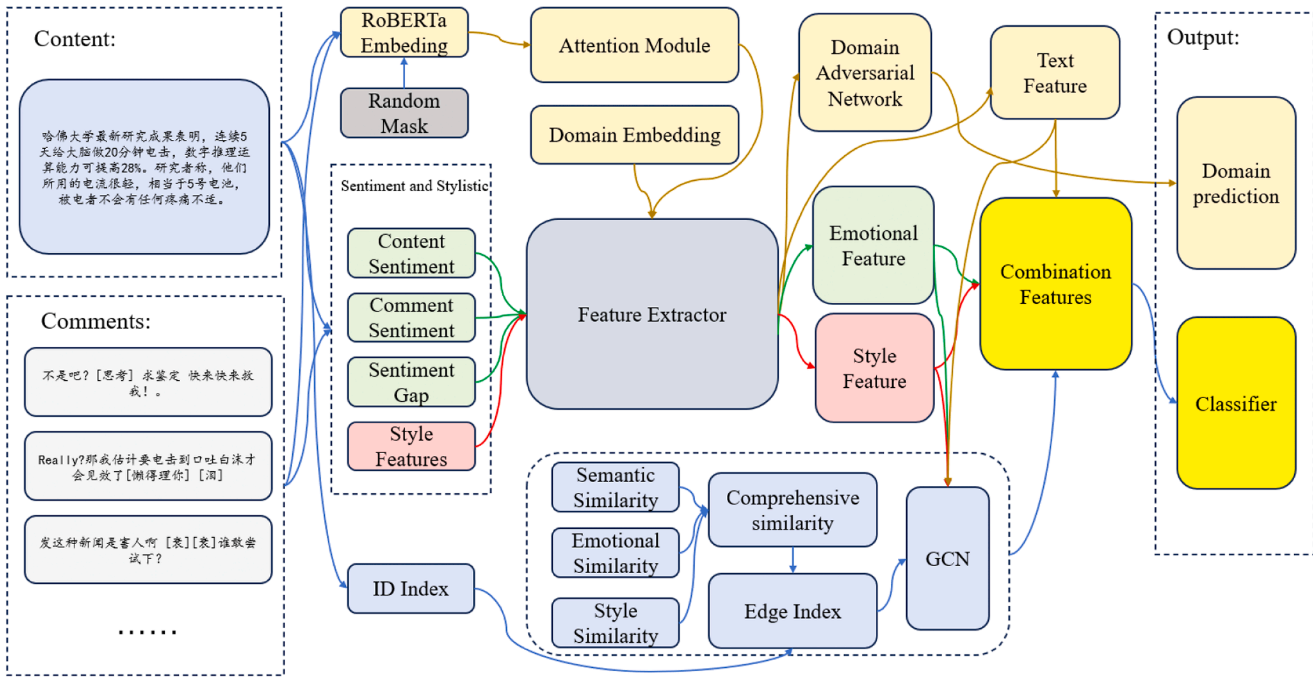

Fig. 1. Overall architecture of MFGAG.

words within the same category; for instance, the emotional weight of "sad," "sorrow," and "anguish" differs, which is similarly reflected in corresponding Chinese terms. Thus, we denote sentiment intensity as D_{int}.

Additionally, natural language contains negation words and degree adverbs. If a negation word appears within the range of the$w$preceding words of the target word, we count its occurrencesneg. For negative sentiments, we consider them to represent the opposite of the corresponding positive emotion; for example, happiness is the opposite of sadness. Degree adverbs multiply the corresponding sentiment intensity by their respective magnitudes, represented as D_{adv}.

Ultimately, the sentiment score for each word can be expressed as follows:

$$E_{t_i} = (-1)^{neg} Emo_{t_i} D_{int} D_{adv} \tag{1}$$

The emotional features of the entire news article are obtained by summing the scores of all words belonging to each sentiment category, followed by concatenating the resulting scores from all categories. This can be expressed mathematically as follows:

$$E_T^e = \sum_{i=1}^{l} E_{t_i}^e, \ \forall e \in E \tag{2}$$

$$E^{word} = E_T^{e_1} | E_T^{e_2} | ...... | E_T^{e_c} \tag{3}$$

In addition to sentiment categories, we also consider the polarity of each word, which can be classified as neutral, positive, or negative. The specific polarity values can be directly obtained from the sentiment lexicon, where the original dictionary represents neutral, positive, and negative sentiments as 0, 1, and 2, respectively. To facilitate algorithmic computation and quantification, we modify the negative polarity to -1. Thus, the emotional polarity for the entire news article can be expressed as follows:

$$E^{pol} = \sum_{i=1}^{l} E_{t_i}^{pol} \tag{4}$$

In the comments section at the bottom left of Fig. 1, we observe a significant presence of emoticons, a phenomenon also common in news articles published on certain self-media or social platforms. Additionally, various personal pronouns and punctuation marks, reflecting different emotional intensities, can reveal users' emotions. Therefore, we conducted a statistical analysis of these elements. By calculating the frequency of their occurrences, we derive an emotional value that contributes to our understanding of the overall sentiment expressed in the comments.

The aforementioned features focus on analyzing emotions from specific, smaller perspectives. To capture the broader emotional characteristics on a larger scale, we manually annotated the sentiment categories for a subset of news articles. Using a multilayer perceptron, we trained a sentiment classifier that concatenates all the previously mentioned features and applies a fully connected layer to obtain the overall sentiment classification for the news article, denoted as E^{cate}. Thus, we combine all emotion-related features to derive the final publisher sentiment E^{pub}:

$$E^{pub} = E^{word} | E^{pol} | E^{aux} | E^{cate} \tag{5}$$

For the sentiment of commenters, which reflects the general emotional perception of the public towards the news, we employed the same methodology used for publisher sentiment. Through average pooling and max pooling, we derived the average sentiment E^{mean} and extreme sentiment E^{max} of the comments. This approach allows us to capture both the general sentiment of news readers as well as the more pronounced emotional reactions, which are crucial for public sentiment analysis. The final representation of the commenter sentiment can thus be expressed as:

$$E^{com} = E^{com}_{mean} | E^{com}_{max} \tag{6}$$

In real-world contexts, certain categories of fake news employ highly subjective language to draw attention and spark emotional engagement, leading to pronounced sentiment alignment among commenters. Conversely, authentic news often encounters skepticism and criticism from its audience. However, in certain domains, the opposite phenomenon occurs entirely. For instance, when citizens are either satisfied or dissatisfied with their country's policies, their emotional responses to genuine versus fabricated news can be diametrically opposed. To examine this phenomenon, we integrate the sentiments of both the publisher and the commenters to derive the sentiment disparity feature. The calculation method for this feature is as follows:

$$E^{gap} = \left(E^{pub} - E^{com}_{mean}\right)\left|\left(E^{pub} - E^{com}_{max}\right)\right. \tag{7}$$

To obtain the final comprehensive emotional feature, we concatenate the three types of sentiment features derived earlier:

$$F^{emo} = E^{pub}|E^{com}|E^{gap} \tag{8}$$

Stylistic feature: We replicated the methodology outlined in paper [29] to process the original dataset, extracting preliminary stylistic features to enhance the existing dataset. This approach allowed us to enrich the dataset with additional stylistic dimensions, facilitating a more comprehensive analysis of the data.

Advanced Features: To conduct a deeper exploration of the preliminary features related to emotion, style, and textual content, we constructed the content extraction network depicted in the Fig. 2 below. The purpose of this network is to perform multi-scale extraction of the foundational features using different convolutional kernels. Specifically, the fourth feature extraction convolutional kernel is defined with a size of 5 × 5.

The process involves the following steps:

Convolutional Layer: Each foundational feature is processed through various convolutional kernels to capture features at multiple scales.

Batch Normalization and Activation: After convolution, batch normalization is applied, followed by the activation function to introduce non-linearity into the model.

1 × 1 Convolutional Kernel: A 1 × 1 convolutional kernel is then applied to refine the extracted features further. This is again followed by batch normalization and activation.

Max Pooling: Max pooling is employed to downsample the feature maps, retaining the most salient features while reducing dimensionality.

Channel Concatenation: The outputs from the five different multi-scale channels are concatenated. However, it's important to note that this is not a simple concatenation of features but rather an integration of extracted insights.

Residual Convolution and Max Pooling: In addition to the above, we apply residual convolutions and residual max pooling to preserve the information from the original input at the maximum scale. The outputs from this process are summed with the multi-scale channels instead of being concatenated.

Linear Transformation: Finally, a linear transformation is applied to adjust the dimensionality of the output before it is produced.

This advanced feature extraction network aims to enhance the richness and depth of the features derived from the initial emotional, stylistic, and semantic analyses.

3.4. Domain adversarial networks

In our previous research and that of others on multi-domain fake news detection, attention mechanisms are commonly employed to achieve accurate domain alignment, which involves correctly identifying the domain to which a news article belongs. Attention mechanisms are particularly effective at uncovering deep relationships within features and assigning varying weights to facilitate domain localization. However, we observed that these mechanisms can sometimes lead to an over-reliance on certain keywords within the text. For example, in military news, terms like "war," "army," and "conflict" tend to receive disproportionately high attention scores. This phenomenon can result in misclassifying other articles containing these words as belonging to the military domain, which can significantly affect the accuracy of fake news detection. As Peng et al. [22] noted, although many fake news detection methods have made some progress, existing approaches almost exclusively rely on global semantic features to detect fake news, neglecting the fact that fake news is semantically inconsistent and difficult to identify using semantics alone.

To address this issue, we adopted an adversarial network approach. For precise domain identification, we prioritize textual content while using attention mechanisms as a supplementary tool. Initially, we introduce random masking after word embeddings to enhance the generalization capabilities of the attention mechanism. Subsequently, a linear layer is employed to compute the attention scores of the input.

$$Scores = Linearr(Embeding(T)) \tag{9}$$

If a position has a mask, we set the attention score for that masked position to negative infinity. This prevents it from contributing to the attention calculations during backpropagation:

$$Scores = Mask\_fill(Scores, -inf)\ if\ mask == 0 \tag{10}$$

By implementing this strategy, we ensure that masked positions are effectively ignored in the attention mechanism, allowing the model to focus solely on the relevant, unmasked features for accurate domain identification and classification.

After completing the text content embedding, we incorporate the corresponding domain embeddings $d$and feed them into the feature extractor depicted in Fig. 2. The extracted content $F^{text}$is then passed to the adversarial network, which processes it through a backward

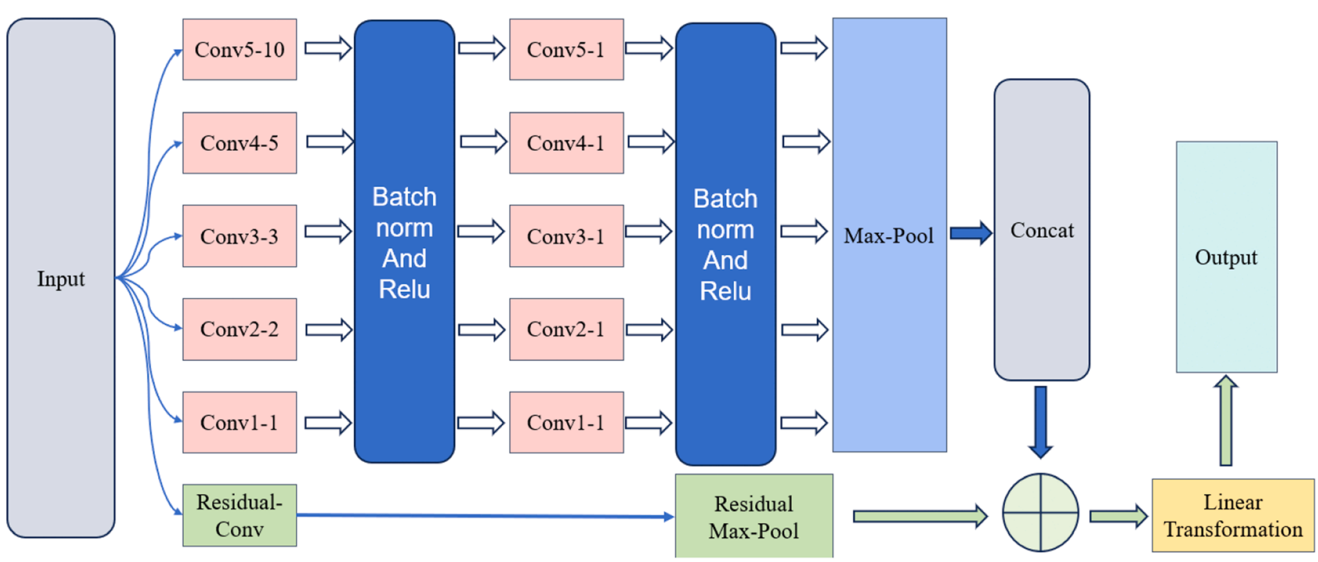

Fig. 2. Advanced feature extractor.

gradient layer and a domain discriminator to produce the domain prediction results:

$$\widehat{F}^{text} = ReverseLayer(F^{text}, \eta) \tag{11}$$

$$d^{pred} = DomainDiscriminator(ReverseLayerF(F^{text})) \tag{12}$$

In this context, η in the backward gradient layer represents the magnitude of the gradient reversal. This mechanism is utilized to balance feature learning and domain adversarial training within the model. By flipping the gradients during backpropagation, we effectively encourage the model to learn domain-invariant features, thereby enhancing its robustness in distinguishing between genuine and fake news across multiple domains.

3.5. Graph convolutional neural networks

In our preliminary research, we observed that certain news events evolve over time, sometimes undergoing further development or even reversal, while also being widely reposted by other media outlets. This results in both temporal and spatial relationships between news items. However, due to the complexity of time-series information—primarily because of dataset limitations over extended time spans—it is challenging to track detailed changes in a single news item over time, particularly in multi-domain news, where data collection is difficult. Therefore, we have opted to focus on uncovering spatial relationships for the time being.

Our research reveals that, within the same time period, a large volume of similar types of fake news often emerges across various formats and categories. These news items may originate from different publishers, and due to differences in stance and perspective, even news about the same event can vary significantly in sentiment, style, and textual features. After precise domain alignment, we still aim to identify distinctive relational patterns between fake and real news.

To achieve this, we designed GCN method to connect these similar news items. By establishing associations among numerous related news pieces, we enhance the accuracy of fake news detection. This approach allows the model to capture deeper interrelations between articles, improving its ability to distinguish fake news through structured, spatially aligned information.

The first step in establishing spatial relationships involves calculating the semantic similarity between two news articles using the textual features F^{text}. From the extracted features, we compute the sentiment similarity from F^{emo} and the stylistic similarity from the style features F^{sty}. The calculation methods are as follows:

$$sim = \frac{F_i \cdot F_j}{||F_i|| \times |F_j \,||} \tag{13}$$

After calculating the individual similarities, we combine them using weighted aggregation to obtain a comprehensive similarity score. This can be expressed as follows:

$$sim\_all = \alpha sim\_text + \beta sim\_emo + \gamma sim\_sty \tag{14}$$

In this context,α,$\beta$and$\gamma$are hyperparameters that will be analyzed and tuned in subsequent experiments.

Next, we will set a threshold ϕ for the comprehensive similarity score. News articles with a similarity exceeding this threshold are considered to exhibit implicit relationships between different categories of fake and real news. We then create edges based on the assigned IDs for each news article, resulting in an adjacency matrix A.

At this stage, we have constructed a spatial graph structure for the news articles, where each node represents a news article and each edge signifies the relational links between them. The feature representation for each node is derived by concatenating the textual features, sentiment features, and style features to form a combined feature vector h_c.

We will then utilize a GCN to process the spatial relationships within the graph:

$$h_{Gcn} = GCN(h_c, A) \tag{15}$$

The GCN will enable us to leverage the interconnectedness of the news articles to enhance our model's ability to discern between fake and real news more effectively. The GCN will perform information passing and aggregation, allowing each node to update its representation based on its neighbors, ultimately improving the detection accuracy of fake news in a multi-domain context.

The resulting features will be concatenated with the combined features to obtain the final classification features. This can be expressed mathematically as:

$$h_z = h_{Gcn} | h_c \tag{16}$$

3.6. News classifier

In the adversarial network, the model outputs the domain prediction results. For the classification of news authenticity, a fully connected layer is employed to classify the final classification features. This can be represented as:

$$y = FC(h_z) \tag{17}$$

The output of this layer will provide the model's predictions regarding the authenticity of the news articles, enabling effective detection of fake news based on the integrated features derived from both the content and the graph structure.

4. Method

In this section, we begin with a comprehensive analysis and description of the dataset, allowing other interested researchers to replicate our experimental results for comparison. Subsequently, we conduct an experimental analysis of the algorithm proposed in this paper from various perspectives. To facilitate readers' understanding, we will outline the experiments conducted to address the following research questions:

RQ1: How effective is the graph network composed of multi-domain news designed in the study?
RQ2: How does the proposed method perform in detecting fake news on multi-domain datasets compared to other multi-domain and single-domain models?
RQ3: What is the impact of various hyperparameters on the performance of the proposed model?
RQ4: What is the contribution of each module in the proposed method to fake news detection?

4.1. Datasets

In this chapter's experiment, we used the Weibo21 dataset created by Nan et al. [3] for algorithm validation. This dataset contains fake and real news snippets collected from December 2014 to March 2021. The dataset includes news articles that have been officially identified as false information by the Weibo Community Management Center[1][1]. Alongside these, real news from the same time periods was also collected, all of which have been verified by NewsVerify[2][2], a platform dedicated to identifying and validating suspicious news on Weibo. Each news article in the dataset includes: (1) the complete and original text of the news content, (2) various modal information, including images, (3) a timestamp providing time-related details, and (4) social reactions, namely

[1] 1 http://service.account.weibo.com/.
[2] 2 https://www.newsverify.com/.

Table 1
Dataset Weibo21.

Domin	Science	Military	Education	Disasters	Politics
Real	143	121	243	185	306
Fake	93	222	248	591	546
all	236	343	491	776	852
	Health	Finance	Entertainment	Society	All
Real	485	959	1000	1198	4640
Fake	545	362	440	1471	4488
all	1000	1321	1440	2669	9128

comments from users. The dataset was obtained by directly contacting the publishers via email. If necessary, you may reach out to the original authors [3] or contact us for access.

The dataset contains 4488 fake news articles and 4640 real news articles, covering nine distinct domains: science, military, education, disasters, politics, health, finance, entertainment, and society. The data has been carefully annotated by several domain experts, and the final statistics are summarized in the figure below:

The figure reveals that the dataset distribution across different domains is unbalanced, which is one of the motivations for introducing the adversarial network: to mitigate the effects of domain bias. Additionally, there are certain disparities in the ratio of fake to real news within each domain, leading to better model performance in some domains and weaker performance in others.

The GCN is employed to capture both intra-domain and inter-domain relationships, which enhances the model's generalization ability. This approach helps alleviate data fragmentation caused by domain-specific alignment. To better address RQ1, we have visualized part of the graph network. Below is a visualization of the global distribution of nodes and edges within the constructed spatial graph. This graph structure illustrates the connections among various news items, providing insights into the relational patterns across the dataset.

Fig. 4 illustrates the connections between different news items in the dataset and how these connections are distributed in the global graph. From nodes with a degree greater than 10, we selected the top 5 nodes with the highest degrees in each category and retained some of their connections. This filtering process highlights representative nodes with high connectivity while simplifying the graph visualization. The bar chart on the right displays the degree distribution of nodes in the global graph, providing insights into the centrality of various news items within the graph structure. The figure demonstrates that relationships between news items are difficult to capture using traditional models. By processing the global graph with a GCN, we can better capture these relationships.

The degree distribution reveals that certain news items may play more influential roles within the dissemination network. These high-degree nodes embody the most prominent features of real and fake news across different domains. By incorporating the adversarial network, we can adjust the model's focus on these influential news items across domains, thus balancing domain bias and domain imbalance within the graph structure. This combined approach enhances the model's ability to generalize across domains while accounting for domain-specific variations in news characteristics.

4.2. Baseline comparison experiment

In the experiments, we address RQ2 by considering three types of baselines, encompassing the following model methodologies:

1. **Single-Domain Baselines**: These include BiGRU [30], TextCNN, and RoBERTa, which are evaluated separately on each domain. The accuracy reported for all data is the average of the accuracies from the nine single-domain experiments.
2. **Mixed-Domain Baselines:** This category includes models like TextCNN, BiGRU, RoBERTa, StyleLSTM [31], and DualEmo [28]. Here, a pruning approach is

(continued on next column)

(continued)

employed, utilizing a single domain extractor for all domains while keeping the other processes consistent
3. **Multi-Domain Baselines:** This includes models such as EANN [32], MMOE [33], MOSE [34], EDDFN [35], MDFEND [3], M3FEND [36] and DCSC [26]. The distinction here is that these models are designed to operate across multiple domains simultaneously, aiming to capture the inter-domain relationships and features

A point that requires special clarification is why we do not compare with large models. This is because research [21] has shown that, although large models demonstrate impressive performance in multi-domain news detection, they have not yet reached the same level of superiority as offline models. The advantages of large models lie in their ability to perform zero-shot detection without training, as well as their more precise understanding of language. However, their limitations include the refusal to answer certain politically sensitive news, particularly regarding platforms like Gemini. This is one of the reasons why large models are excluded from the comparison. Furthermore, Li et al. [21] compared large models with MDFEND [3], and ChatGPT-3.5 achieved an accuracy of around 0.86. Even the latest GPT-4o has an average accuracy of only 0.885 across eight domains in the dataset. Another reason for not using large models is that most state-of-the-art large models are internet-connected, while the dataset we use contains news from 2021, which has already been verified on various websites. As a result, an online large model could simply retrieve results rather than employing reasoning. This makes a direct comparison with offline models unfair. The more equitable comparison would involve using the most recent fake news, but unfortunately, we do not have the capability to continuously access the latest fake news data. This limitation is beyond our ability to address. Li et al. [21] used news from 2023 to 2024 and continuously integrated the latest news to ensure that it reflects the changing nature of misinformation and real-time events, but there are only 335 real and fake news in total.

This structured comparison allows us to assess the effectiveness of different strategies in handling domain-specific and cross-domain challenges in fake news detection.

Table 2.

From the experimental results presented in Table 2, it is evident that our proposed method achieved optimal results in most domains. Although we did not secure the best performance in the scientific, military, society, and entertainment domains, we ranked within the top three in each. In the five domains where we performed optimally, we achieved an average accuracy improvement of 3.3 percentage points compared to the best results from other studies, which represents a significant enhancement. Through our analysis of the intermediate modules in the algorithm, we found that in the science and military domains, as discussed in Section 3.4, the heavy reliance on domain-specific terminology or less commonly used words results in a significant dependence on domain keywords. Consequently, the suppression of certain keywords may have an adverse effect, leading to performance degradation, while other domains benefit from such suppression. In contrast, in the social and entertainment domains, our analysis of the graph network revealed substantial overlap with other domains. Specifically, news in these two domains often forms connections with news from numerous other specialized domains. Upon further analysis, we found that the inclusion of stylistic features caused many non-specialized news articles from other domains to be misclassified as social and entertainment news, resulting in a complex and diverse set of articles. Although these articles are mostly fake news, the incorporation of adversarial networks mitigated some of the negative impact, maintaining the model's performance near the top, despite a slight decrease in accuracy.

Additionally, our performance metrics on the mixed-domain dataset indicated optimal results, with average improvements of 0.8 percentage points in F1 score, accuracy (ACC), and area under the curve (AUC),

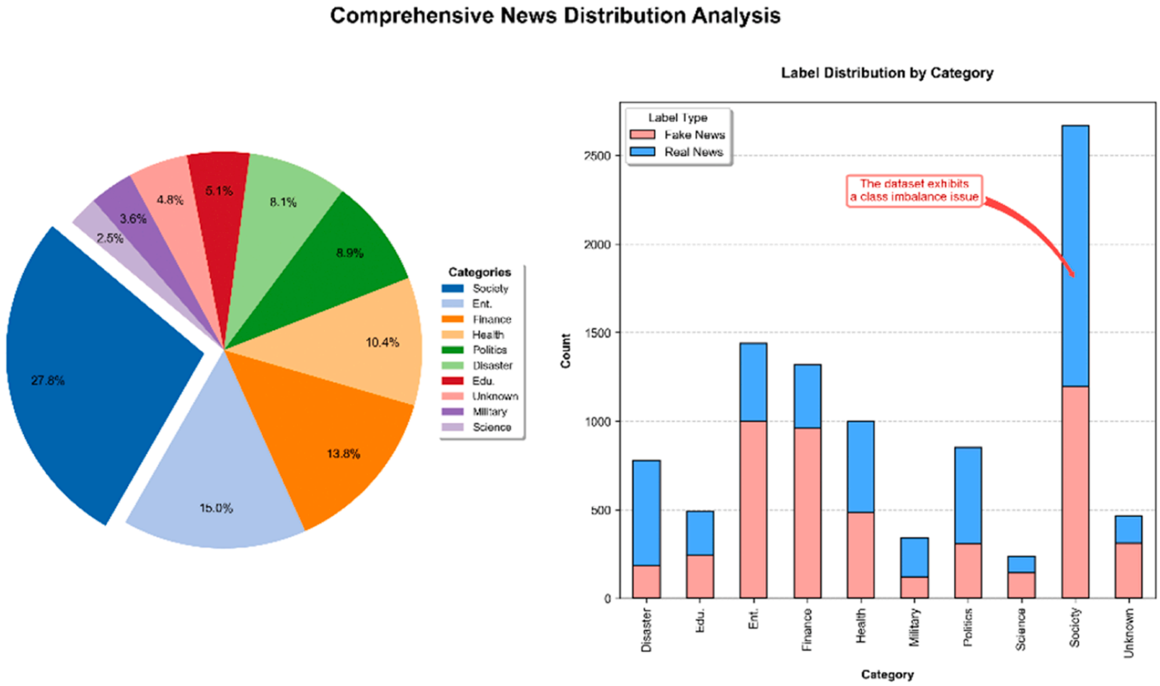

Fig. 3. Field and label distribution analysis.

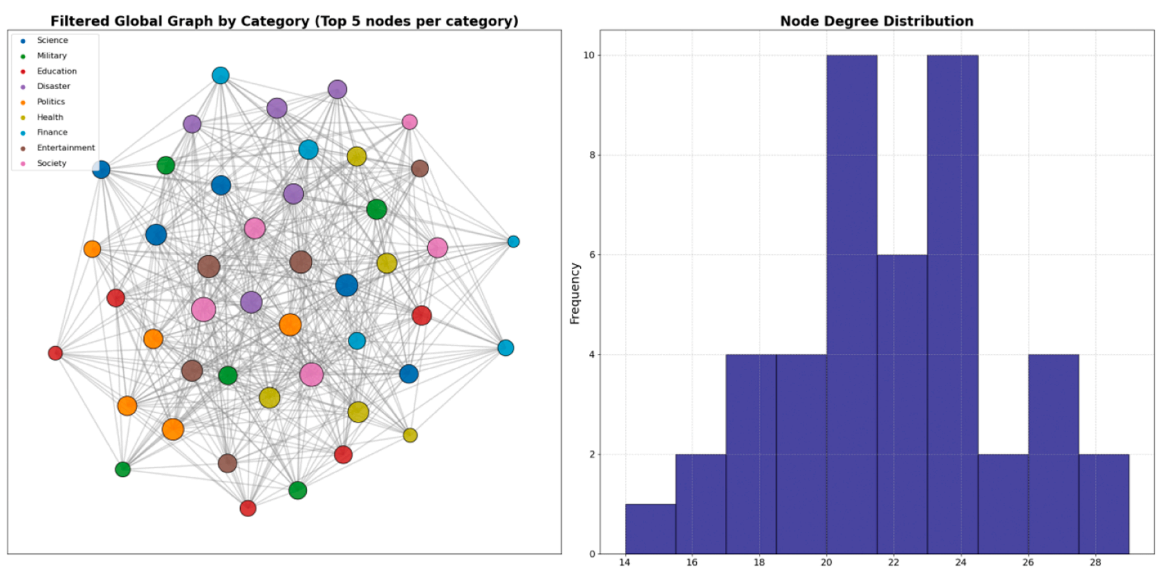

Fig. 4. Global distribution of nodes and edges.

which is overall Accuracy, overall AUC and overall F1-score in Table 2. These findings demonstrate that the integration of graph neural networks and adversarial networks, built upon the foundational features, effectively enhances multi-domain recognition and precise localization in fake news detection.

To provide a clearer, more intuitive depiction of our algorithm, we visualized the validation and test-set performance across different training epochs under cross-validation (see Fig. 5). Additionally, We independently conducted 20 runs of cross-validation, recording the Accuracy, AUC, and F1-score for each run. A paired sample *t*-test was then performed to evaluate the differences between these performance metrics. The purpose of the paired *t*-test is to compare two measurement methods by assessing the differences in their results under the same experimental conditions—each pair of data points corresponds to measurements taken from the same run. The results are summarized in Table 3.

4.3. Hyperparameter experiments

In this section, to provide a more thorough answer to RQ3, we conduct experiments on the hyperparameters involved in the proposed model to explore the deeper factors affecting model performance.

First, we examine the gradient reversal coefficient η in the domain adversarial network, which controls the magnitude of the reversed gradient in the gradient reversal layer. In domain adaptation tasks, the choice of flip coefficient is very critical. Suppose we have a source domain and a target domain. When training a model, we usually hope that the features of the source domain can be well transferred to the

Table 2
Statistical analysis of experimental performance.

Model	Type	Science	Military	Education	Disasters	Politics	Health
BiGRU	single	0.518	0.337	0.742	0.729	0.859	0.837
TextCNN	single	0.407	0.337	0.806	0.439	0.848	0.882
RoBERTa	single	0.746	0.737	0.815	0.755	0.804	0.887
BiGRU	mixed	0.727	0.872	0.814	0.794	0.836	0.887
TextCNN	mixed	0.725	0.884	0.836	0.822	0.856	0.877
RoBERTa	mixed	0.778	0.907	0.833	0.851	0.837	0.909
StyleLSTM	mixed	0.773	0.919	0.834	0.853	0.849	0.880
DualEmo	mixed	0.832	0.903	0.836	0.840	0.846	0.891
EANN	multi	0.823	0.927	0.862	0.867	0.871	0.915
MMoE	multi	0.876	0.911	0.871	0.877	0.892	0.936
MoSE	multi	0.85	0.886	0.882	0.867	0.881	0.918
EDDFN	multi	0.819	0.914	0.868	0.879	0.848	0.938
MDFEND	multi	0.83	0.939	0.892	0.9	0.887	0.94
M3FEND	multi	0.829	**0.951**	0.9	0.89	0.883	0.946
DCSC	multi	**0.931**	0.85	0.893	0.89	0.916	0.918
MFGAG(Ours)	multi	0.849	0.942	**0.94**	**0.938**	**0.923**	**0.975**

Model	Type	Finance	Ent.	Society	overall Acc	overall AUC	overall F1
BiGRU	single	0.814	0.799	0.792	0.81	0.89	0.81
TextCNN	single	0.822	0.797	0.862	0.837	0.909	0.837
RoBERTa	single	0.836	0.851	0.83	0.848	0.923	0.848
BiGRU	mixed	0.829	0.863	0.849	0.86	0.931	0.86
TextCNN	mixed	0.864	0.846	0.854	0.869	0.938	0.869
RoBERTa	mixed	0.874	0.877	0.858	0.88	0.945	0.88
StyleLSTM	mixed	0.88	0.885	0.855	0.882	0.947	0.882
DualEmo	mixed	0.905	0.894	0.857	0.885	0.954	0.885
EANN	multi	0.871	0.896	0.888	0.898	0.961	0.898
MMoE	multi	0.857	0.889	0.875	0.895	0.955	0.895
MoSE	multi	0.867	0.891	0.873	0.894	0.954	0.894
EDDFN	multi	0.864	0.883	0.887	0.892	0.953	0.892
MDFEND	multi	0.895	0.907	0.898	0.914	0.971	0.914
M3FEND	multi	0.901	**0.932**	0.909	0.922	0.975	0.922
DCSC	multi	0.906	0.918	**0.924**	0.921	0.977	0.921
MFGAG(Ours)	multi	**0.908**	0.923	0.911	**0.931**	**0.981**	**0.931**

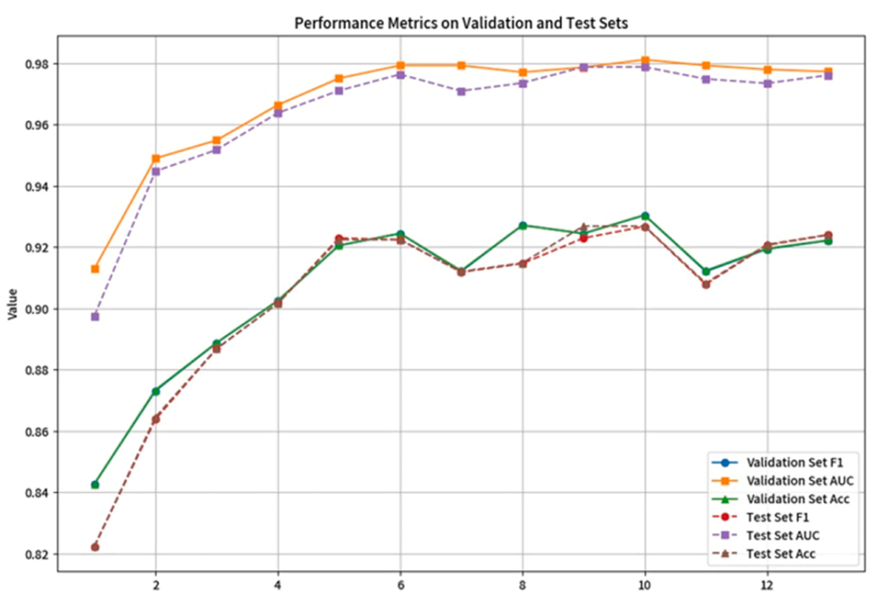

Fig. 5. Performance metrics on validation and test sets.

Table 3
Results of the *t*-tests.

	t	p
ACC vs AUC	−55.49	1×10^{-19}
ACC vs F1	2.95	9×10^{-3}
AUC vs F1	54.79	1.23×10^{-19}

target domain. The gradient flip layer forces the model to not simply rely on the features of the source domain by reversing the gradient of the source domain features, but to learn to share more universal features between the source and target domains. We tested several values for η, ranging from 0.005 to 1, and compared the resulting performance of the model. This analysis allows us to assess the impact of different gradient reversal strengths on the overall accuracy and generalization capability of the model.

As shown in Table 4, by adjusting the value of the reversal coefficient η, we can control the strength of domain adaptation. η larger value encourages the feature extractor to "resist" the domain classifier's judgments more strongly, promoting the learning of features that are invariant across domains. This helps prevent the model from relying on less significant features and "cutting corners". The reversal coefficientηalso allows us to balance the weights between the source task and the target domain, ensuring that the model learns effectively in the source domain while adapting well to the target domain.

Through our experiments, we discovered that performance peaks when the value is set to 0.5, indicating that the gradient magnitude is reduced to half of its original size. This suggests that during adversarial training, the model experiences weaker adversarial influences compared to when the gradient is not diminished. We posit that, as previously mentioned, in the scientific and military domains, excessively strong adversarial settings can lead to significant performance degradation, thereby corroborating our analysis presented in Table 2. Consequently, we believe that a coefficient of 0.5 allows the model to train without being overly "penalized," effectively balancing the stability of training with the benefits of adversarial optimization. This balance prevents the model from making overly aggressive adjustments that could cause it to focus excessively on certain domains, thereby artificially inflating accuracy in these areas and ultimately impairing performance in other domains. This phenomenon is indicative of one aspect of the model's lazy optimization. Our experimental results also demonstrate that a moderate gradient reversal enhances the model's generalization capabilities.

Subsequently, we conducted experiments on the hyperparameters α, β, and γused to compute the overall similarity in constructing the graph network. This helped us explore which type of similarity—semantic, emotional, or stylistic—should have a greater influence on the determination of the domain and the identification of fake news. The results indicate that setting α =0.5, β =0.3, and γ =0.2 yielded the highest F1 score. This confirms that semantic information plays a dominant role in fake news detection. After analyzing various parameter combinations and their corresponding results, we concluded that sentiment and stylistic features exhibit a notable degree of commonality across news articles from multiple domains. In most instances, these features serve to distinguish between fake and genuine news. During the algorithm design process, we prioritized the algorithm's ability to differentiate between domains, with semantic features maintaining a dominant role in this regard. Consequently, we incorporated semantic features separately into the adversarial network. However, we cannot determine the causal relationship behind this decision—specifically, whether the separate integration of semantic features into the adversarial network inherently increased their significance. To address this uncertainty, we conducted an isolated analysis in Experiment 4.4.

Moreover, when comparing sentiment features to stylistic features, sentiment features proved to be relatively more important. As previously mentioned, the sentiments expressed in comments on fake and genuine news are contrasting across different domains. This opposing nature of sentiment makes it more critical for accurate domain determination than stylistic features.

The impact of adjusting different parameters on the final results indirectly supports the answer to RQ1, demonstrating that structuring multi-domain news into a graph network yields a highly objective classification performance.

Table 4
gradient reversal coefficient η.

η	F1	Acc	AUC
0.005	0.915	0.915	0.976
0.01	0.918	0.918	0.975
0.1	0.922	0.922	0.976
0.25	0.917	0.917	0.975
0.5	**0.931**	**0.931**	**0.981**
0.75	0.927	0.927	0.979
1	0.916	0.922	0.958

Next, we explored the threshold $\phi$for overall similarity, determining the level of similarity needed between neighbouring nodes to establish a relational link between them. As shown in Fig. 7, the results indicate that setting the threshold at 0.7 maximized the algorithm's performance. Beyond this point, performance dropped sharply as the threshold increased.

When the threshold deviates from this value—either falling below or exceeding it—performance declines rapidly. Through targeted analysis, we selected certain nodes to examine the subnetworks formed at different thresholds. We observed that when the threshold is set above 0.7, the network links become increasingly sparse, preventing a fully connected graph structure and hindering information exchange among some nodes, which in turn degrades performance. Conversely, when the threshold falls below 0.7, reducing it further only increases the number of links. Because both sentiment and semantic similarities are factored into the threshold calculation, even unrelated news items retain some degree of similarity in these two aspects, resulting in a relatively high minimum similarity threshold. This explains why a high threshold of 0.7 delivers the best performance.

Additionally, we noted an unexpected phenomenon: real and fake news across multiple domains tend to cluster in certain threshold intervals. Concretely, some subnetworks form isolated "islands" disconnected from other nodes, resembling community detection or clustering. This observation further supports the effectiveness of our method for constructing the graph network.

4.4. Adversarial network module input experiment

In our hyperparameter experimentation involving the gradient reversal coefficient $\eta$for the adversarial network, we noted that the relationship between this coefficient and performance may be influenced by the nature of the input data. In Experiment 4.3, as shown in Fig. 1, only the processed semantic features were fed into the adversarial network. To conduct a deeper analysis, we then combined sentiment and stylistic features with the semantic features and re-examined the effect of varying η. The results of this extended investigation are presented in the Table 5 below:

From the data in Table 5, it is evident that feeding only semantic features into the adversarial network yields the best results. Our analysis suggests that semantic features encompass the most comprehensive domain-discriminative information, and applying a suitably calibrated gradient reversal prevents the model from being excessively "penalized." This approach maintains a balance between training stability and adversarial optimization, while also mitigating the lazy optimization phenomenon caused by overreliance on certain features.

In contrast, sentiment features only help determine domains in some instances and can negatively impact overall performance when used in a multi-domain mixed dataset. Lastly, as our previous analysis indicated, including stylistic features decreases performance. When treating a given domain as the source domain to compare against other domains, stylistic differences are not pronounced enough to be beneficial. Under the same training parameters, adding these features merely increases the model's confusion, resulting in lower performance. Nevertheless, we anticipate that performance could improve again with an increased number of training epochs or a larger dataset.

4.5. Ablation experiment

In response to RQ4, we conducted ablation experiments on the modules designed in the model, specifically focusing on the adversarial network, graph neural network, and the residual convolution in feature extraction. The results presented in the table illustrate how the overall performance of the model changes when each corresponding module is

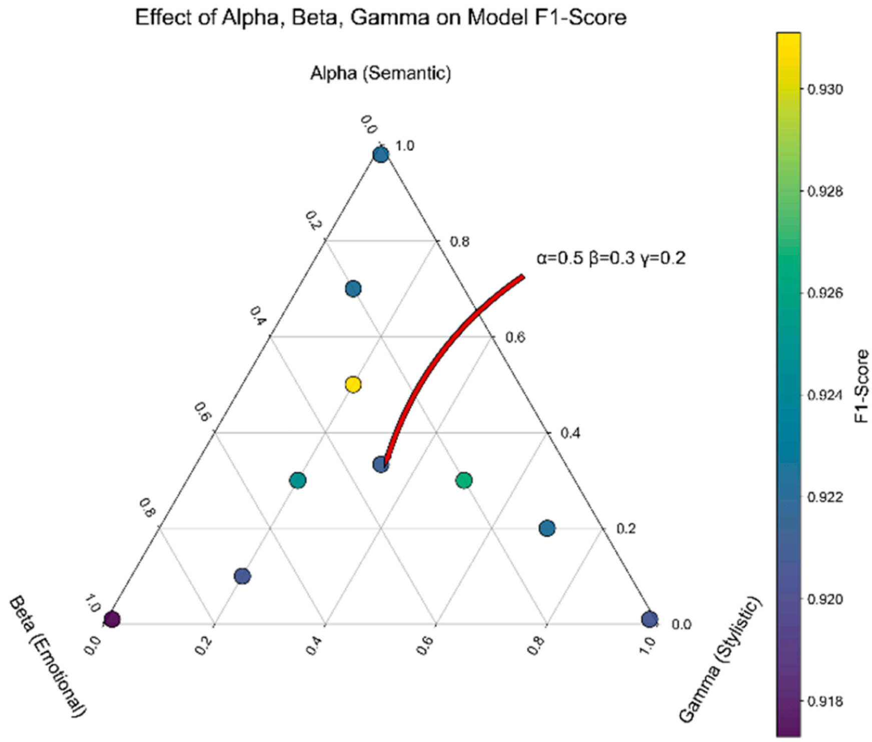

Fig. 6. Effect of α,β,γon Model F1-Score.

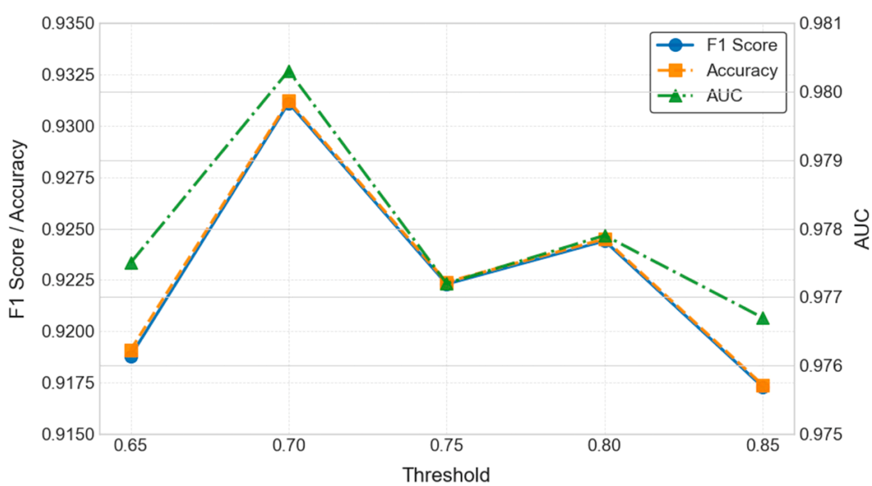

Fig. 7. Threshold $\phi$and model performance metrics.

Table 5
Module ablation.

Input	η	F1	Acc	AUC
Semantic Feature	0.1	0.922	0.922	0.976
	0.5	**0.931**	**0.931**	**0.981**
	1	0.916	0.922	0.958
Semantic and Emotional Feature	0.1	0.921	0.919	0.974
	0.5	0.93	0.932	0.978
	1	0.914	0.922	0.961
Semantic and Stylistic Feature	0.1	0.911	0.916	0.958
	0.5	0.922	0.919	0.954
	1	0.925	0.924	0.965
Semantic, Emotional and Stylistic Feature	0.1	0.924	0.922	0.966
	0.5	0.927	0.93	0.976
	1	0.921	0.915	0.951

removed. This analysis highlights the contribution of each module to the model's effectiveness, demonstrating the crucial role these components play in enhancing performance for fake news detection across multiple domains.

As shown in Table 6, the comprehensive module configuration yields the optimal performance. In addition, as anticipated, the residual convolution module has the most significant impact on performance, as

Table 6
Module ablation.

Ablation module	F1	Acc	AUC
Complete Model	0.931	0.931	0.98
Domain Adversarial Networks	0.924	0.924	0.975
Graph Convolutional Neural Networks	0.929	0.929	0.979
Residual Convolution Block	0.923	0.923	0.977

all data extraction and mining are conducted through this module. Following this, the adversarial network module plays a crucial role in enhancing the accuracy of domain classification. Lastly, the graph network has the minimal influence. Our analysis indicates that relying solely on textual similarity for analysis limits the performance of the graph network. Moreover, the essential multi-hop mechanism within the graph network has limited effectiveness in fake news detection, as the semantic relevance between news items separated by two hops is minimal. This often leads to the incorporation of information from other domains, thereby adversely affecting performance. Therefore, introducing additional modal information is necessary to achieve further improvements.

5. Conclusion

In conclusion, this research presents a comprehensive framework for detecting fake news across multiple domains by integrating advanced techniques such as adversarial networks and graph neural networks. Our approach innovatively combines sentiment analysis, stylistic features, and semantic embeddings to enhance the robustness and accuracy of fake news detection. The use of adversarial training mitigates domain bias, while graph-based representations effectively capture relationships between news articles, improving generalization across diverse categories.

This study addresses critical challenges such as domain transfer and data distribution discrepancies, advancing the theoretical understanding of fake news detection—challenges that have long hindered the scalability of detection models across multiple domains. Unlike existing research, which either examines each domain independently or focuses on generalized content, this work uniquely explores the spatial correlations of news across diverse domains, establishing a holistic approach to relational studies. By integrating adversarial networks with graph neural networks, we propose an innovative framework capable of not only identifying the authenticity of news articles but also unveiling the intricate interrelations between articles within shared temporal contexts. The multidimensional approach, combining sentiment, style, and semantic features, establishes a new paradigm for feature extraction and domain alignment in multi-domain fake news detection. These insights deepen the theoretical foundation for constructing robust models capable of handling the complexities of misinformation across diverse domains. From a practical perspective, this research enhances the reliability and applicability of fake news detection systems in real-world scenarios. The proposed framework demonstrates significant improvements in accuracy, achieving an average increase of 3.3 percentage points in single-domain settings and nearly 1 percentage point in mixed-domain datasets, with an overall accuracy reaching 93 %. This performance boost underscores the model's ability to generalize effectively across domains, making it a valuable tool for mitigating the spread of misinformation. Moreover, the use of graph neural networks for relational modeling and adversarial training for domain alignment ensures adaptability to dynamic and diverse news ecosystems. This makes the proposed approach particularly suitable for applications in media monitoring, social platforms, and public opinion management. Building upon the current work, we plan to incorporate the visual modality present in fake news to further enhance the accuracy of domain classification. While existing multimodal approaches primarily utilize images for the purpose of fake news detection, their potential for improving domain-level classification remains largely unexplored. Addressing this gap presents a significant and highly meaningful challenge for future research.

Future research will focus on further refining the model by incorporating temporal dynamics to better track the evolution of news narratives over time. Additionally, exploring more sophisticated methods for feature extraction and leveraging larger datasets will be crucial to enhance the model's performance and adaptability in real-world scenarios. Expanding the framework to include multilingual capabilities could also broaden its applicability and effectiveness in diverse linguistic contexts.

CRediT authorship contribution statement

Xuefeng Li: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Jian Wei:** Methodology, Formal analysis, Data curation. **Chensu Zhao:** Visualization, Investigation. **Xiaqiong Fan:** Supervision, Funding acquisition. **Yuhang Wang:** Validation, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Our algorithms and data are all publicly available on GitHub and are subject to peer verification. We also introduce the original dataset and the acquisition method in detail in the article.

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